Calibration-Free Edge-Computing IMC Macro With Direct Current-to-Digital Conversion

Andrea Fasolino, Rosalba Liguori, *Member, IEEE*, Luigi Di Benedetto, *Senior Member, IEEE*, Alfredo Rubino, *Graduate Student Member, IEEE*, and Gian Domenico Licciardo, *Senior Member, IEEE*

***Abstract*—In-memory computing (IMC) has emerged as a promising solution to the "memory wall" problem of traditional Von Neumann architectures by integrating computation directly within the memory. This brief presents a novel current-mode IMC macro that leverages a nanoampere-range temperature-independent reference current and direct current-to-digital conversion. The proposed design mitigates power inefficiencies and thermal instability of previous architectures without the need for calibration. Implemented in TSMC LP 65 nm CMOS technology, the design achieves an energy efficiency of 310.7 TOPS/W and an area efficiency of 18.93 TOPS/mm^2. The reference current generator ensures a temperature coefficient of just 363 ppm/°C over a temperature ∈ [-10; 115]° C. Using a VGG-6 model on the CIFAR-10 dataset with 87.93% accuracy, the drop between the software baseline model and IMC hardware (1bA/1bW/5bO) at TT@27°C/FF@115°C/SS@-10°C is 0.17/0.32/0.51%.**

***Index Terms*—Calibration-free, current source, edge computing, in-memory computing, multiply-accumulate, thermal stability, ultra-low power.**

I. Introduction

IN-MEMORY computing (IMC) has emerged as a promising solution to overcome the limitations of traditional Von Neumann architectures [1], where separate memory and processing units result in a bottleneck known as the "memory wall" [2]. IMC aims to solve this issue by merging the processing functions within the memory element.

The Multiply-Accumulate (MAC) operation is ubiquitous in modern applications, such as neural networks, digital signal processing, etc.: $D_{MAC} = \sum_{i=0}^{M-1} IA_i \times \omega_i$ where IA_i and ω_i are the i-th input and weight. Depending on how the digital MAC value (D_{MAC}) is computed, IMC can be divided into digital-based [3] and analog-based [4], [5], [6], [7], [8], [9], [10], [11] approaches. Digital IMC performs accumulation and dimension reduction using digital logic [3]. The robustness of this approach and its ease of integration make it a strong candidate for modern spatial architectures. However, the high overhead of full-precision digital circuits limits their energy and area efficiency. Analog IMC performs MAC operation in the analog domain and mainly attains three operational modes: charge-, time-, and current-based. The charge-based IMC performs multiplication using the charge on capacitors and accumulates via charge redistribution. Thanks to well-controlled capacitor parameters, charge-domain IMC offers high robustness to PVT variations and accuracy [7], although it necessitates larger bitcells (BCs) compared to other approaches. The time-based IMC operates through the control of a delay unit [8]. Compared to current- and charge-based methods, it offers a larger sense margin but with greater latency. Time-based computation is more efficient because of its lower toggle rate. However, like the current-based approach, it suffers from PVT variations that reduce accuracy, requiring PVT tracking techniques to mitigate delay fluctuations. The current-based IMC [9] leverages a current $I_{MAC} = D_{MAC} \times I_{ref}$ where the I_{ref} value determines the performance of the whole macro.

Previous current-based IMC macros [9], [10], [11] focus on finding an optimal balance between throughput and computational accuracy since these are the main disadvantages when compared to less energy-efficient digital-based IMC macros [3]. Furthermore, careful process, supply voltage, and temperature (PVT) tracking is necessary for two main reasons: (I) to mitigate current fluctuations that could lead to incorrect results; (II) to calibrate the Analog-to-Digital converters (ADCs). Reference [9] proposes a 6T BC with PMOS access transistors, which mitigates the read disturb issue attributed to the conventional 6T-SRAM BC with a single-slope 5b ADC, where 32x128 BCs are used for ADC calibration. Reference [10] leverages dual 7T BC, enabling the encoding of inputs on 3, 7, or 15 levels with BCs reconfigurability. Here, 48 BCs are used for offset calibration and 96 BCs for binary-searching ADC. The current-mode IMC macro in [11] exploits a process and temperature auto-compensation scheme, composed of a binary-weighted resistor DAC with a PVT-tracking feedback transistor.

In order to lower the power consumption and the two-step conversion overhead, a current-mode IMC macro exploiting 13 nA reference current (I_{ref}) and a direct current-to-digital conversion from I_{MAC} to D_{MAC} is proposed in this brief. Moreover, since current-mode IMC heavily suffers from temperature variations, and given the need to calibrate the ADCs, a temperature-independent reference current source is proposed, leveraging the combination of Low Voltage Threshold (LVT) and High Voltage Threshold (HVT) Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET), resulting in a Constant with Temperature (CWT)

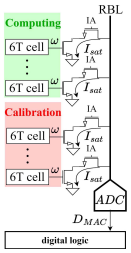
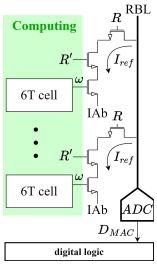
Received 26 March 2025; revised 30 May 2025; accepted 17 June 2025. Date of publication 19 June 2025; date of current version 1 August 2025. This brief was recommended by Associate Editor S. Taneja. *(Corresponding author: Gian Domenico Licciardo.)*

The authors are with the Department of Industrial Engineering, University of Salerno, 84084 Fisciano, Italy (e-mail: afasolino@unisa.it; rliguori@unisa.it; ldibenedetto@unisa.it; arubino@unisa.it; gdlicciardo@unisa.it).

Digital Object Identifier 10.1109/TCSII.2025.3581384

1549-7747 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:32:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PREVIOUS CURRENT-MODE IMC MACROS WITH CALIBRATION BCS VS THE PROPOSED CALIBRATION-FREE ARCHITECTURE

	Previous [9]–[11]	**This Work**
Type	current - 8T SRAM	current - 9T SRAM
Example		
Pros	• high frequency • smaller BC	• low power consumption • direct current-to-digital ADC • calibration-free
Cons	• BCs for calibration • limited power eff. due to I_{sat} two step ADC	• lower speed due to parasitic capacitance charge/discharge with small current

I_{ref}. This ensures low Differential Non-Linearity (DNL) and Integral Non-Linearity (INL) without the need for calibration. The main contribution of this brief can be summarized as follows (Table I):

- power consumption reduction thanks to nano amperes reference current;
- direct current-to-digital IMC macro that provides higher density and even lower power consumption, resulting in higher energy efficiency;
- CWT current reference, to avoid thermal variability of reference current, needing no calibration for the ADC.

The proposed circuit is designed with Cadence Virtuoso in TSMC CMOS low-power (LP) 65 nm technology. Post-layout simulations at various corners and temperatures show that the worst DNL and INL are under 0.5 LSB, with an energy efficiency of 310.7 TOPS/W and an area efficiency of 18.93 TOPS/mm^2. The temperature coefficient of the reference current is just 363 ppm/°C over a temperature range of −10 to 115 °C. On actual ML loads, using a VGG-6 model on the CIFAR-10 dataset, the worst accuracy drop between the software model and IMC hardware (1bA/1bW/5bO) at normal operating conditions (TT@27°C) is 0.87%.

II. SUB-COLUMN OPERATION

A sub-column operation consists of the partial product generation operated by each 9T BC and the current accumulation and current-to-digital conversion operated by the LCB.

A. 9T BC

The proposed BC (Fig. 1) is composed of (I) the standard 6T circuit, made of M_0, M_1, and the two cross-coupled inverters, (II) the MOSFET M_p that executes the $IA \times \omega$ product, (III) the high impedance branch (HIB) of a cascode current mirror, made of M_3 and M_4, that replicates the current I_{ref}.

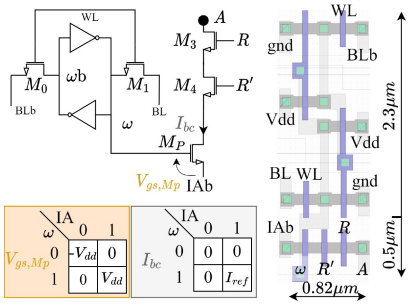


Fig. 1. Proposed 9T BC schematic and layout in TSMC CMOS LP 65 nm.

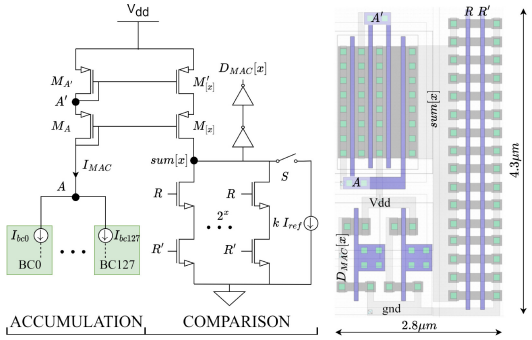


Fig. 2. Proposed local computing block (LCB) with the layout of the MSB computation circuit in TSMC CMOS LP 65 nm. The pitch is matched with the BC to ensure high density.

The proposed 9T BC executes both conventional and IMC operations. The conventional 6T SRAM R/W operation is executed through M_0 and M_1, activated by the WL. If it is necessary to use the IMC macro as a standard SRAM memory, the I_{ref} generator can be disabled, thus having $V(R) = V(R') = gnd$. When the I_{ref} generator is enabled, in the branch composed by M_3, M_4, and M_P will flow a current only if $V_{GS,M_P} > V_{TH,M_P}$: when $\omega = IA = 1$ it is $V_{GS,M_P} = V_{dd}$, thus the BC current $I_{bc} = I_{ref} \simeq 13\ nA$. This approach allows high control over I_{bc}, enabling nanoampere-range currents for the computation. The main drawback is that node A presents high-value capacitance that must charge/discharge with a current proportional to I_{ref}. Thus, although the power consumption can be strongly reduced, the design must cope with a limited maximum operating frequency. The BC leakage current $I_{off} \simeq 5\ pA$ flows when M_P is off (worst condition), leading to $\frac{I_{ref}}{I_{off}} \simeq 2.6\ 10^3$. This does not affect the MAC precision since at most $I_{MAC} = 128\ I_{off} = 0.64\ nA << I_{ref}$.

B. Local Computing Block

All the I_{bc} are summed in the node A, resulting in I_{MAC} that flows in M_A and $M_{A'}$ (Fig. 2). I_{MAC} is mirrored through the cascode current mirror composed by M_A, $M_{A'}$, $M_{[x]}$, and $M'_{[x]}$ to compute each bit of D_{MAC}. Thanks to proper sizing of the HIB transistors, I_{MAC} is downscaled of 4×, resulting in $0 \leq I_{MAC} \leq 31\ I_{ref}$. The HIB of this cascode current mirror is replicated for each bit of D_{MAC} (from $D_{MAC}[4]$ to $D_{MAC}[0]$ in this brief). Each HIB is connected to the node sum[x], where x corresponds to the bit of D_{MAC}. The pull-down network is designed to try to pull a current $I^*_{[x]} = \sum_{i=x}^{X-1} D_{MAC}[x]\ 2^x\ I_{ref}$

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:32:20 UTC from IEEE Xplore. Restrictions apply.

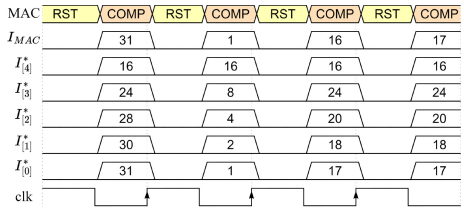


Fig. 3. Time diagram of a MAC computation in the proposed LCB. The currents are scaled to I_{ref}. The macro is reset at the rising edge of the clock, while the computation starts at the falling edge.

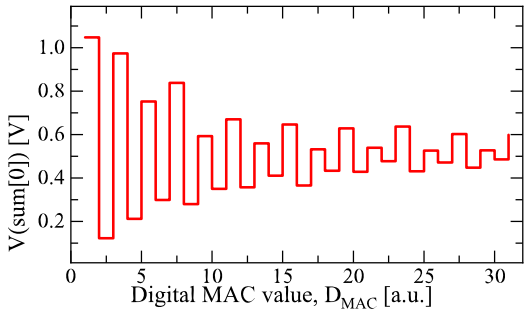


Fig. 4. Voltage at node sum[0] corresponding to the LSB comparator node vs digital MAC value D_{MAC}.

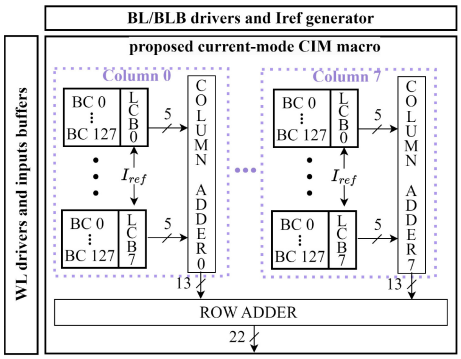


Fig. 5. Proposed IMC macro: 8 columns with 8 sub-columns, each containing 128 BCs and 1 LCB.

from the node sum[x]. Where X is the number of bits of D_{MAC} ($X = 5$ in this brief). To implement this strategy, the comparison block of each bit has 2^x parallel HIBs that produce a current equal to 2^x I_{ref}, and a switch S enabled from LSBs of D_{MAC}.

Fig. 3 illustrates an example of the values of each $I^*_{[x]}$ for various I_{MAC}. During the computation phase, the inputs are multiplied and accumulated, whereas in the reset phase, the macro is disabled, and the voltage at all nodes is brought to ground. The computation phase can have different duty cycles. Lowering the duty cycle of this phase reduces the consumed energy. In this example, the duty cycle of the computation phase is equal to 50%, thus, the duty cycle of the reset phase is (100-50)%. When $I_{MAC} = 31$ I_{ref}, all the output bits of D_{MAC} are asserted; thus, the comparison currents are 16/24/28/30/31 I_{ref}. When $I_{MAC} = 1$ I_{ref} only $D_{MAC}[0]$ is asserted, thus the comparison currents are the minimum (16/8/4/2/1 I_{ref}, respectively). Other two examples are provided with $I_{MAC} = 16$ I_{ref} and $I_{MAC} = 17$ I_{ref}. Because in both cases $D_{MAC}[4]$ is asserted, all the $I^*_{[x]}$ for $x < 4$ are incremented of 16 I_{ref}, but in the second case, the last comparator will give an asserted output, resulting in $D_{MAC}[0] = 1$.

Equation (1) describes the ideal behavior of the current comparator circuit.

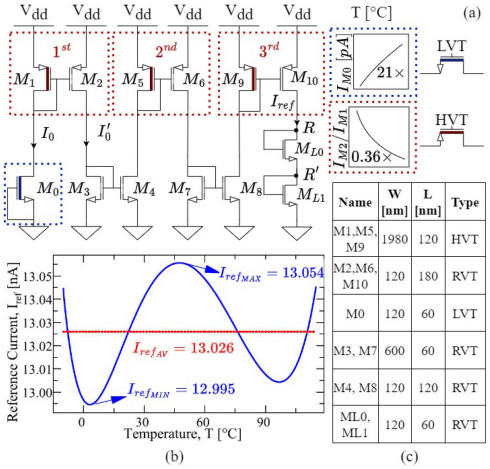
$$V(sum[x]) \simeq \begin{cases} 0 \quad for \quad I_{MAC} < I^*_{[x]} \\ V_{dd} \, for \quad I_{MAC} \geq I^*_{[x]} \end{cases} \tag{1}$$

The series of $M'_{[x]}$ and $M_{[x]}$ has higher impedance than the pull down network when $I_{MAC} < I^*_{[x]}$, thus $V(sum[X])$ is hold to ground, since $V(R') > V_{dd} - V(A')$ and $V(R) > V_{dd} - V(A)$. When $I_{MAC} \geq I^*_{[x]}$, the current that flows in the branch is equal to $I^*_{[x]}$. V(X') and V(X) reach their maximum value, thus $V(X') \leq V_{dd} - V(S')$ and $V(X) \leq V_{dd} - V(S)$, resulting in $V(sumX) \neq 0$. With this methodology, the current flowing in each comparison block is given by $min(I_{MAC}, I^*_{[x]})$. Since $D_{MAC}[0]$ presents the highest commutation rate and smallest signal margin (SM), as expected for the least significant bit (LSB) of an ADC, in Fig. 4 the behavior of its voltage against the corresponding D_{MAC} value is shown. The SM in a conventional 5-bit ADC for a supply voltage of 1.2 V is equal to 38.7 mV, and it is uniform over all the input range, whereas the worst SM in the proposed design is 41.53 mV, which corresponds to the difference in the voltage at node sum[0] between $D_{MAC} = 29$ and $D_{MAC} = 30$. The proposed direct current-to-digital approach improves the signal margin of a conventional ADC by 7%. Considering headroom limitation due to the stacking of the devices, the critical nodes are A and sum[x] of Fig. 2. Due to the drops on M_A and M'_A (diode-connected MOSFETs), the minimum voltage value in A must be in the range [0.65, 0.5] V, depending on the temperature [−10, 115] °C. The voltage at node sum[x] depends on the activated branches as shown in Fig. 4.

III. Proposed IMC Macro

The proposed IMC macro (Fig. 5) is composed of Word Line (WL) drivers and IA buffers, BL/BLB drivers, digital column and row adders, 9T SRAM computing array, and I_{ref} generator. The computing array consists of one row and eight columns, each containing 8 sub-column elements (128 BC and 1 LCB). The sub-column operation provides 128 MACs with 1bIA/1bW/5bO. The result of each sub-column operation can be taken as is or can be merged with the others in the same column, producing a column result of 128 MACs with 1bIA/8bW/13bO. Moreover, there is the possibility to merge all the columns' results in the row adder, producing an 8bIA/8bW/22bO final result column, and row adders are made of digital circuits that properly shift and sum the inputs to take into account their positional value. The proposed current reference (Fig. 6a) leverages three main building blocks: (I) LVT MOSFET operating in subthreshold mode ($V_{gs}(M_0) = 0$); (II) dual VT PMOS current mirror with an HVT diode-connected PMOS and an RVT PMOS on the HIB; (III) RVT NMOS current mirror. The LVT MOSFET subthreshold current is given by [12]. When drain-to-source voltage $V_{ds} > 4 \times V_t$ the effect of V_{ds} can be neglected. Since V_{TH,M_0} has a linear negative dependence on

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:32:20 UTC from IEEE Xplore. Restrictions apply.



Name	W [nm]	L [nm]	Type
M1,M5, M9	1980	120	HVT
M2,M6, M10	120	180	RVT
M0	120	60	LVT
M3, M7	600	60	RVT
M4, M8	120	120	RVT
ML0, ML1	120	60	RVT

Fig. 6. (a) Proposed CWT current reference, with the thermal characteristics of the sub-threshold LVT NMOS stage and the dual-VT PMOS stage. (b) Reference current I_{ref} vs temperature T. The Temperature Coefficient (TC) is equal to 363 ppm/°C for a temperature $T \in [-10, 115]$ °C. (c) MOSFETs' dimensions in the current reference generator.

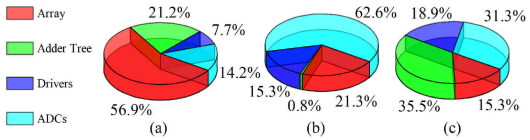


Fig. 7. Breakdown of (a) area occupation, (b) power consumption in 1bW/1bIA/5bO mode, and (c) power consumption in 8bW/8bIA/22bO mode.

T, whereas V_T increases, the overall trend of $I_{M_{LVT}}$ is a positive exponential against temperature. The negative dependence of the mirrored current of the dual VT PMOS current mirror $\alpha_P(T) = \frac{I_0'}{I_0}$ compensates for this behavior. Three PMOS stages are used with the same current ratio $\alpha_P(T)$ and two NMOS stages with current ratio α_N. $\alpha_P(T)$ is used to obtain the target temperature behavior, whereas α_N to set the average value I_{refAV}. Analytical values for transistors' width (W) and length (L) are obtained, and then a simulation results-based fine-tuning is operated, leading to the values reported in Fig. 6c. The proposed generator supplies an $I_{refAV} = 13$ nA with a Temperature Coefficient (TC) equal to 363 ppm/°C over a temperature range from −10 to 115 °C (Fig. 6b). This calibration-free reference feeds the circuit through the voltage at nodes R and R', in the range [631; 369] mV and [301; 174] mV, respectively, for T ∈ [−10; 115] °C.

IV. Results and Comparison

The results presented are obtained through Cadence Virtuoso post-layout simulations with TSMC CMOS LP 65 nm technology libraries. The whole IMC macro occupies an area of 191 × 127 μm^2 for 8 Kb memory, without any BC for ADC compensation or trimming. The breakdown of area occupation, power consumption in 1bW/1bIA/5bO, and in 8bW/8bIA/22bO are depicted in Figs. 7 (a), (b), and (c), respectively. Fig. 8 shows the behavior of D_{MAC} against I_{MAC}/I_{ref} with various technology corners, namely Slow NMOS Slow PMOS (SS), Fast NMOS Fast PMOS (FF), Typical NMOS Typical PMOS (TT), at different temperatures.

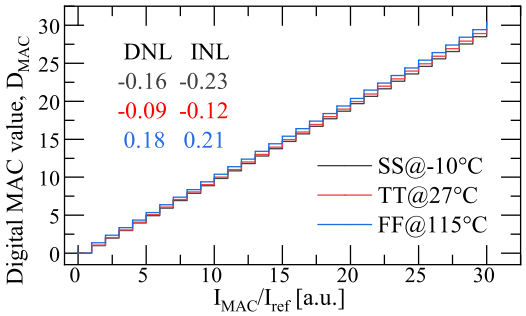


Fig. 8. MAC current I_{MAC} scaled to I_{ref} vs digital MAC value D_{MAC}, INL, and DNL at −10/27/115 °C for SS/TT/FF technology corner, respectively.

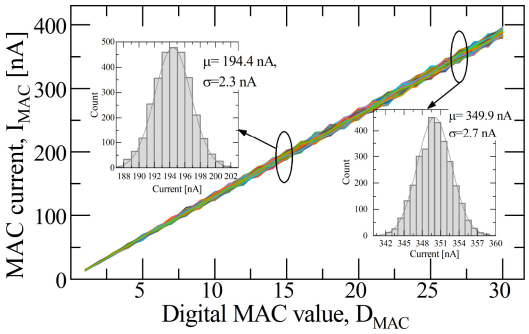


Fig. 9. Linearity of MAC current I_{MAC} vs digital MAC value D_{MAC} with 3000 Monte Carlo simulations.

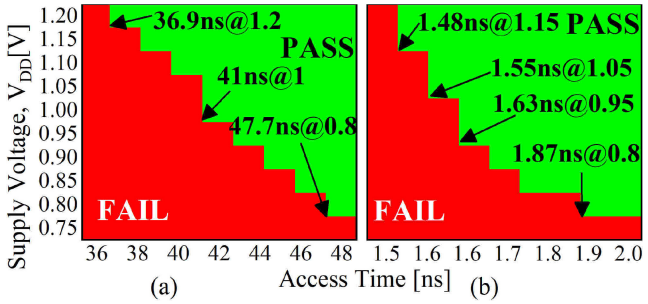


Fig. 10. Simulated Shmoo plot of V_{DD} versus compute latency in (a) IMC in 1bIA/1bW/5bO configuration, (b) SRAM array used as a standard memory.

These cases represent the worst scenarios (SS@-10°C and FF@115°C) and the normal operation scenario (TT@27°C). The maximum DNL is 0.18 LSB FF@115°C, and the maximum INL is −0.23 LSB SS@-10°C, both below 0.5 LSB. Fig. 9 depicts the MAC current against D_{MAC} obtained with 3000 Monte Carlo runs. The insets show the histogram distribution for the worst relative error is 1.18% for $D_{MAC} = 15$, and the maximum absolute error is 2.7 nA for $D_{MAC} = 27$. Fig. 10 depicts the Shmoo plot obtained through post-layout simulations. Fig. 10a presents IMC mode in 1bIA/1bW/5bO configuration, Fig. 10b SRAM array used as a standard memory storage macro.

Table II presents accuracy comparisons of various ML models on MNIST and CIFAR-10 datasets. These results are obtained through simulations, including the HW properties when executing each arithmetic operation of the models across different process-temperature conditions (TT@27°C, FF@115°C, SS@-10°C), with the mapping scheme presented

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:32:20 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SIMULATIONS WITH HW PROPERTIES OF ACTUAL ML LOADS

	Dataset	Model	SW	IMC HW		
Bit-precision			1bA/1bW/5bO			
Condition				TT@27°C	FF@115°C	SS@-10°C
Accuracy [%]	MNIST	MLP	96.25	95.56	95.39	95.28
		ResNet-18	98.60	97.73	97.61	97.44
	CIFAR-10		84.21	83.88	83.72	83.37
		VGG-6	87.93	87.82	87.61	87.42

TABLE III
COMPARISON WITH STATE-OF-THE-ART SRAM-BASED IMC MACROS

	Proposed	[3]	[7]	[9]	[10]	[11]
Tech [nm]	65	55	55	65	65	65
BC	9T	6T	11T-1C	6TP	Dual 7T	8T
Mode	Current	Digital	Charge	Current	Current	Current
Array [bit] (ADCs)	128x64 (0x64)	64x64 (0x64)	256x64 (0x64)	96x128 (32x128)	528x128 (144x128)	128x64 (0x64)
Density** [Kb/mm^2]	**337.7**	66	4.08	327.7	130	10.45
Vdd [V]	1.2	0.7-1.2	0.125-1	1.2/0.67	0.45/0.8	1
W/IA/O [bit]	1/1/5	1/1/7	1/1/8	1/1/5	1/1.5/6	4/4/4
E. Eff.*** [TOPS/W]	**310.7**	2238	111	157.6	258.5	23.76
A. Eff.*** [TOPS/mm^2]	**18.93**	13.21	0.34	12.16	10.4	1.06
Results	Post-layout	Meas.	Meas.	Post-layout	Meas.	Post-layout
Acc. [%] (MNIST/ CIFAR-10)	97.73/ 87.82	97.13/ 82.45	NA/ 90.03	96.5/ 87.2	97.62/ 87.99	NA/ 95.02

*: Energy Efficiency (E. Eff.), Area Efficiency (A. Eff.), Measurements (Meas.), Technology node (Tech.), and Accuracy (Acc.)
**: no calibration BCs are included.
***: efficiencies refer to the W/IA/O configuration reported in the table.

in [10]. A binary sigmoid activation is employed as a nonlinear function. The models utilized are a multi-layer perceptron (MLP) [10], the ResNet-18, and the VGG-6, properly modified to handle the input data. The worst accuracy drop between the software model and IMC hardware (1bA/1bW/5bO) at normal operating conditions (TT@27°C) is 0.87%. Table III presents the results of the proposed macro compared to the state-of-the-art. The proposed macro achieves the literature cell density, considering only the BC actually used in the MAC computation. The throughput is $2 \times 64 \times 128/36.9 = 444$ GOPS in a 1bA/1bW/5bO configuration. This leads to Energy (FF@115/TT@27/SS@-10) and Area Efficiency of 293.6/310.7/327.1 TOPS/W and 18.93 TOPS/mm^2, respectively.

V. CONCLUSION

This brief presents a novel current-mode IMC macro with good computational accuracy, energy, and area efficiency over a wide temperature range, without calibration. Future developments will explore the support for higher precision, as well as the integration into a complete edge-AI system.

REFERENCES

[1] A. Fasolino et al., "Multiclass object classification using ultra-low resolution time-of-flight sensors," *IEEE Sensors Lett.*, vol. 8, no. 10, pp. 1–4, Oct. 2024.

[2] W. Sun et al., "A survey of computing-in-memory processor: From circuit to application," *IEEE Open J. Solid-State Circuits Soc.*, vol. 4, pp. 25–42, 2024.

[3] H. You, W. Li, D. Shang, Y. Zhou, and S. Qiao, "A 1–8b reconfigurable digital SRAM compute-in-memory macro for processing neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 4, pp. 1602–1614, Apr. 2024.

[4] S. Kim et al., "DynaPlasia: An eDRAM in-memory computing-based reconfigurable spatial accelerator with triple-mode cell," *IEEE J. Solid-State Circuits*, vol. 59, no. 1, pp. 102–115, Jan. 2024.

[5] S. Kim et al., "Scaling-CIM: eDRAM in-memory-computing accelerator with dynamic-scaling ADC and adaptive analog operation," *IEEE J. Solid-State Circuits*, vol. 59, no. 8, pp. 2694–2705, Aug. 2024.

[6] Z. Xuan, C. Liu, Y. Zhang, Y. Li, and Y. Kang, "A brain-inspired ADC-free SRAM-based in-memory computing macro with high-precision MAC for ai application," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 4, pp. 1276–1280, Apr. 2023.

[7] Y. Lin et al., "An 11t1c bit-level-sparsity-aware computing- in-memory macro with adaptive conversion time and computation voltage," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 11, pp. 4985–4995, Nov. 2024.

[8] P.-C. Wu et al., "An 8b-precision 6T SRAM computing-in-memory macro using time-domain incremental accumulation for AI edge chips," *IEEE J. Solid-State Circuits*, vol. 59, no. 7, pp. 2297–2309, Jul. 2024.

[9] I. Shah, K. Sarfraz, and M. Chan, "An efficient 6tp SRAM-based CIM macro with column ADCS for binarized neural networks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 5, pp. 2509–2513, May 2024.

[10] C. Yu, H. Jiang, J. Mu, K. T. C. Chai, T. T.-H. Kim, and B. Kim, "A dual 7T SRAM-based zero-skipping compute-in-memory macro with 1-6b binary searching ADCS for processing quantized neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 8, pp. 3672–3682, Aug. 2024.

[11] H. Kim, Y. An, R. Kim, S. Kim, and Y. Shim, "Union SRAM: PVT variation auto-compensated, bit precision configurable current mode 8T SRAM in memory MAC macro," *IEEE Access*, vol. 12, pp. 162882–162893, 2024.

[12] R. De Rose, F. Crupi, M. Lanuzza, and D. Albano, "A physical unclonable function based on a 2-transistor subthreshold voltage divider," *Int. J. Circuit Theory Appl.*, vol. 45, no. 2, pp. 260–273, 2017.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:32:20 UTC from IEEE Xplore. Restrictions apply.